

Supplementary Material 1: Methods and Hyper-parameters Used to Generate Layouts

Table 1 contains the list of methods and hyper-parameters used for each of the layouts shown in the paper.

Figure	NLDR method	Hyper-parameter(s)
1a	UMAP	n_neighbors = 30, min_dist = 0.3
1b	UMAP	n_neighbors = 5, min_dist = 0.8
1c	UMAP	n_neighbors = 5, min_dist = 0.01
1d	tSNE	perplexity = 5
1e	tSNE	perplexity = 30
1f	PHATE	knn = 5
1g	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
1h	PaCMAP	n_neighbors = 30, init = random, MN_ratio = 0.9, FP_ratio = 5
2	tSNE	perplexity = 47
4a	tSNE	perplexity = 47
5b	tSNE	perplexity = 47
6	tSNE	perplexity = 47
8a	tSNE	perplexity = 47
8b	tSNE	perplexity = 62
8c	UMAP	n_neighbors = 15, min_dist = 0.1
8d	PHATE	knn = 5
8e	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
8f	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2
10a	UMAP	n_neighbors = 30, min_dist = 0.3
10b	UMAP	n_neighbors = 5, min_dist = 0.8
10c	UMAP	n_neighbors = 5, min_dist = 0.01

(continued)

Figure	NLDR method	Hyper-parameter(s)
10d	tSNE	perplexity = 5
10e	tSNE	perplexity = 30
10f	PHATE	knn = 5
10g	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
10h	PaCMAP	n_neighbors = 30, init = random, MN_ratio = 0.9, FP_ratio = 5
11a	UMAP	n_neighbors = 30, min_dist = 0.3
11e	tSNE	perplexity = 30
12a	tSNE	perplexity = 30
12b	tSNE	perplexity = 89
12c	UMAP	n_neighbors = 15, min_dist = 0.1
12d	PHATE	knn = 5
12e	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
12f	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2
13a	tSNE	perplexity = 30
14a	tSNE	perplexity = 30
A4a	tSNE	perplexity = 71
A4b	UMAP	n_neighbors = 15, min_dist = 0.1
A4c	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2
A5	tSNE	perplexity = 52
A6a	UMAP	n_neighbors = 30, min_dist = 0.3
A6b	tSNE	perplexity = 30
A7a	UMAP	n_neighbors = 30, min_dist = 0.3
A7b	tSNE	perplexity = 30
A8a	UMAP	n_neighbors = 30, min_dist = 0.3
A8b	tSNE	perplexity = 30
A9a	UMAP	n_neighbors = 30, min_dist = 0.3
A9b	UMAP	n_neighbors = 5, min_dist = 0.8
A9c	UMAP	n_neighbors = 5, min_dist = 0.01
A9d	tSNE	perplexity = 5
A9e	tSNE	perplexity = 30
A9f	PHATE	knn = 5

(continued)

Figure	NLDR method	Hyper-parameter(s)
A9g	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
A9h	PaCMAP	n_neighbors = 30, init = random, MN_ratio = 0.9, FP_ratio = 5
A10a	tSNE	perplexity = 30
A10b	tSNE	perplexity = 89
A10c	UMAP	n_neighbors = 15, min_dist = 0.1
A10d	PHATE	knn = 5
A10e	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
A10f	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2

Table 1: NLDR methods and hyper-parameters used for each Figure in the main paper.